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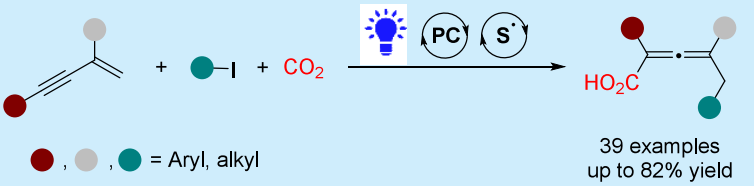
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Photocatalytic 1,4-Carbocarboxylation of 1,3-Enynes with CO_2

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- CO_2 utilization
- Tetrasubstituted 2,3-allenoic acid
- Mild reaction conditions
- Efficient regio- and chemoselectivity
- High functional group tolerance
- Drug late-stage modification

ABSTRACT: Herein, we report a photocatalytic regioselective 1,4-carbocarboxylation of 1,3-enynes with CO_2 enabled by a thiol-mediated hydrogen atom transfer (HAT) and formate cooperative mechanism. This strategy uses formate and CO_2 as synergistic carbon sources, to deliver a diverse array of aryl-, heteroaryl-, and alkyl-functionalized tetrasubstituted allene carboxylic acids. The reaction exhibits broad substrate scope across electronically and sterically varied enynes and iodides, and facilitates late-stage functionalization of bioactive molecules. Gram-scale synthesis and derivatization experiments further demonstrate its synthetic utility.

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Allenes, characterized by their unique cumulated diene system, are widely found in natural products and bioactive molecules (Scheme 1a),[1] and represent highly versatile building blocks in modern organic synthesis and drug design. The radical-mediated approach has emerged as a key strategy for allene synthesis, providing efficient, diverse, and controllable routes to these structures. Its notable advantages include facilitating multifunctionalization and enabling the construction of complex molecular architectures.[2] Recent advances have established several effective methods for generating allene intermediates, such as 1,4-additions to 1,3-enynes,[2a,3] activation of propargylic derivatives via leaving groups,[2a,b,3a,4] and HAT from preformed allenes[5] have been effectively employed to generate allene intermediates (Scheme 1b). The resulting intermediates exhibit broad reactivity, engaging with electrophiles, nucleophiles, and radical coupling partners, thus providing efficient pathways to tetrasubstituted allenes and significantly promoting the synthesis and application of allene-derived compounds.

Among these, multifunctionalized tetrasubstituted 2,3-allenoic acids are of particular interest due to their axially chiral scaffolds and notable bioactivity, demonstrating considerable potential in medicinal chemistry and functional materials.[6] The carboxylic acid group not only enhances bioavailability but also provides a versatile handle for further derivatization, attracting broad attention from synthetic chemists.[1a,7] Conventional synthetic routes to such compounds typically involve the oxidation of allenols or hydrolysis of allene nitriles and esters (Scheme 1c),[8] which often suffer from drawbacks including multistep sequences and poor atom economy. Earlier strategies relied on stoichiometric metal reagents to generate allenyl metal intermediates, followed by carboxylation with CO_2.[9] Although transition-metal-catalyzed carbonylative reactions using CO allow the introduction of carboxylic acid groups, these methods require high-pressure conditions and expensive metal catalysts.[6,10]

More recently, electrocatalytic approaches have been developed for the conversion of propargylic derivatives with CO_2 into 2,3-allenoic acids under mild conditions.[4a,b] However, these methods are currently limited to the synthesis of monofunctionalized products. Therefore, the efficient construction of multifunctionalized tetrasubstituted allenoic acids remains a significant and unresolved challenge in synthetic chemistry.

Carbon dioxide, serving as a renewable and abundant C1 synthon, offers a direct route to high-value carboxylic acids in alignment with green chemistry principles.[11] Nevertheless, its high thermodynamic stability and kinetic inertness present

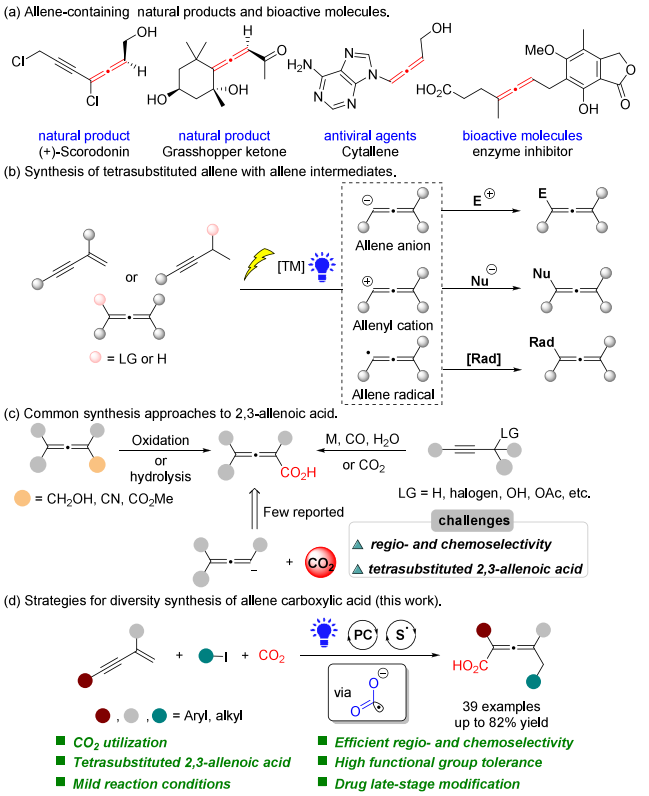
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Scheme 1. Literature Reports and Our Work



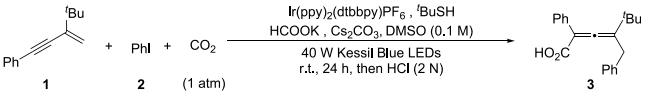
substantial challenges for catalytic carboxylation. In this context, photocatalytic carboxylation has emerged as a promising strategy for activating simple substrates, owing to its operational simplicity, sustainability, and cost efficiency.[12] Of particular note is the photochemical generation of highly reactive carbon dioxide radical anions ($CO_2^{\bullet-}$), which has attracted considerable attention for driving efficient carboxylation reactions under mild conditions.[13]

Inspired by our group's previous achievements in CO_2 functionalization,[14] we sought to develop a method for synthesizing bifunctionalized tetrasubstituted allene carboxylic acids via the regioselective 1,4-difunctionalization of 1,3-enynes. To overcome the persistent challenge of competitive radical 1,2-additions in such transformations,[2a,3a,15] we designed a dual-role photocatalytic system. This approach employs thiol as a HAT catalyst, along with formate acting both as a precursor for $CO_2^{\bullet-}$ radicals and an auxiliary carbon source. By engaging a broad range of aryl, heteroaryl, and alkyl iodides, this strategy enables highly selective access to diverse difunctionalized tetrasubstituted 2,3-allenoic acids.

To validate our hypothesis, we employed enyne **1** as the model substrate to investigate the 1,4-carbocarboxylation of 1,3-enynes (Table 1). Systematic optimization of the reaction conditions

revealed that employing $Ir(ppy)_2(dtbbpy)PF_6$ as the photocatalyst, *tert*-butyl thiol as the hydrogen atom transfer (HAT) catalyst, cesium carbonate as the base, and potassium formate as the reductant afforded the target allene carboxylic acid **3** in 82% isolated yield with excellent chemoselectivity (entry 1; see Supporting Information (SI) for detailed optimization). The reaction did not proceed in the absence of blue light irradiation (entry 2), other control experiments confirmed the necessity of all components as omitting photocatalyst, thiol, formate, or

Table 1. Reaction Condition Optimization[a]



Entry	variation from the above conditions	yield (%)[b]
1	none	88 (82)
2	In dark	N.D.
3	w/o PC/^{t}BuSH/HCO_2K	N.D.
4	w/o CO_2 (N_2 atmosphere)	31
5	4DPAIPN instead of $Ir(ppy)_2(dtbbpy)PF_6$	52
6	*fac*-$Ir(ppy)_3$ instead of $Ir(ppy)_2(dtbbpy)PF_6$	N.D.
7	DABCO instead of ^{t}BuSH	73
8	HCO_2Cs instead of HCO_2K	73
9	K_2CO_3 instead of Cs_2CO_3	67
10	DMF instead of DMSO	24
11	PhBr instead of PhI	53
12	$PhBF_3K$ instead of PhI	20

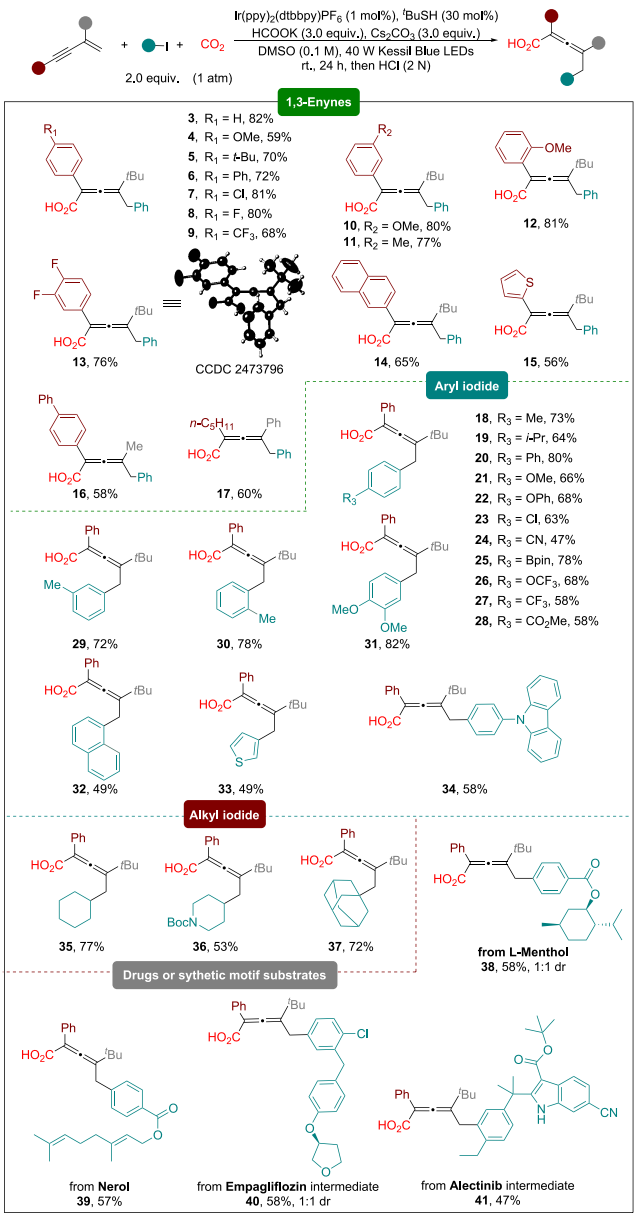
[a]Reaction conditions: **1** (0.2 mmol), **2** (0.4 mmol), 1 atm CO_2, $Ir(ppy)_2(dtbbpy)PF_6$ (1 mol %), ^{t}BuSH (0.1 mmol), HCOOK (0.6 mmol), Cs_2CO_3 (0.6 mmol), DMSO (2 mL), irradiation with 40 W Kessil Blue LEDs at room temperature (rt) for 24 h, then 2 N HCl. [b]Yields were determined by ^{1}H NMR using dibromomethane as an internal standard, and the isolated yield was given in parentheses.

base abolished product generation (entry 3). Experiments conducted under a N_2 atmosphere in the absence of CO_2 resulted in a significant decrease in product yield (Entry 4), clearly demonstrating that gaseous CO_2 is an essential component of the reaction. Increasing the equivalent amount of formate had limited effect on yield improvement (see SI), indicating that formate primarily functions as a reductant, with only minor and inefficient contribution as a carbon source. Substitution of $Ir(ppy)_2(dtbbpy)PF_6$ with 4DPAIPN diminished the yield to 52% (entry 5), while other Ir-photocatalysts exhibited lower efficiency (entry 6). Evaluation of alternative HAT catalysts revealed reduced yields with DABCO (73%) (entry 7). Comprehensive screening of alternative formates, bases, and solvents uniformly resulted in diminished yields (entries 8−10). Other radical precursors, namely both bromobenzene and potassium benzotrifluoroborate, exhibited markedly reduced reactivity compared to iodobenzene, resulting in substantial recovery of starting material. (entries 11−12).

After carefully screening various reaction parameters, the substrate scope of the 1,4-carbocarboxylation reaction involving 1,3-enynes and iodides was systematically investigated under the optimized conditions (Scheme 2). Enyne substrates bearing electron-donating (methoxy, *tert*-butyl), neutral (phenyl), and electron- withdrawing (fluoro, chloro, trifluoromethyl) groups all afforded the desired products in moderate to good yields (B), with no significant difference in reactivity observed (**3**-**9**). Both ortho- and meta-substituted enynes provided comparable yields (**10**−**12**). Difluoro-substituted, fused-ring and heterocyclic enynes were also smoothly converted (**13**−**15**). Furthermore, the structure of product **13** was unambiguously determined by X-ray diffraction analysis (CCDC 2473796). In addition to *tert*-butyl-substituted enynes, methyl-substituted enynes were also compatible with this reaction, providing the desired product in moderate yield (**16**). Furthermore, alkyl-substituted enynes underwent analogous transformation to afford the target product in 60% yield (**17**).

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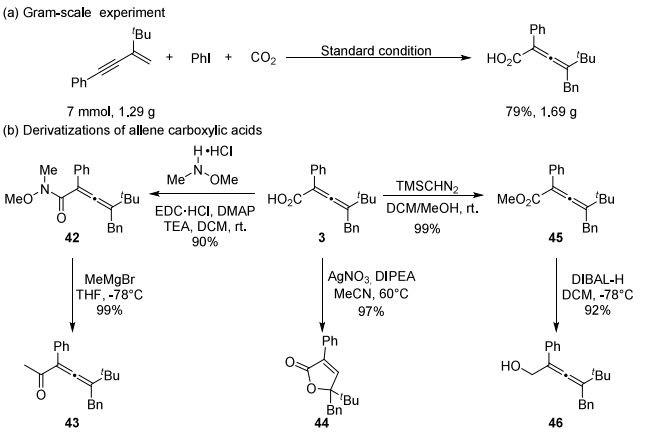
Scheme 2. Substrate Scope Investigation



To further broaden the scope of this methodology, we explored the applicability of iodides bearing diverse substituents and substitution patterns. Evaluation of the substrate scope revealed excellent compatibility with aryl iodides containing neutral (methyl, isopropyl, phenyl), electron-donating (methoxy, phenoxy), and electron-withdrawing (chloro, cyano, boronate ester, trifluoromethoxy, trifluoromethyl, ester) groups (**18**−**28**, 47%−80%). Nevertheless, strongly electron-withdrawing substituents led to markedly diminished reaction efficiency, as reflected in substantially lower yields (**24**, **27**). Aryl iodides with ortho or meta substituents and disubstituted aryl iodides both maintained good yields (**29**−**31**), indicating high tolerance to steric hindrance. Fused-ring iodides (**32**), sulfur-containing heterocyclic iodides (**33**), and nitrogen- containing heterocyclic iodides (**34**) all afforded moderate yields, demonstrating good heteroatom compatibility. In addition to aryl iodides, the methodology was successfully extended to alkyl iodides, as exemplified by the efficient conversion of secondary iodocyclohexane (**35**), *N*-Boc-4-iodopiperidine (**36**), and tertiaryiodoadamantane (**37**). In contrast, primary alkyl halides exhibited limited reactivity under the current conditions. The method exhibited excellent applicability for the late-stage modification of complex molecules, as iodinated derivatives of L- Menthol, Nerol, Empagliflozin intermediate, and Alectinib intermediate all provided the target carboxylated products in moderate yields (**38**−**41**), highlighting its utility for the late-stage functionalization of bioactive molecules.

To evaluate the synthetic potential of this reaction, enyne **1** was subjected to gram-scale conditions (7 mmol), affording the allene carboxylic acid product **3** in 79% isolated yield (1.69 g) (Scheme 3a). Serving as a versatile synthetic building block

Scheme 3. Synthetic Utility



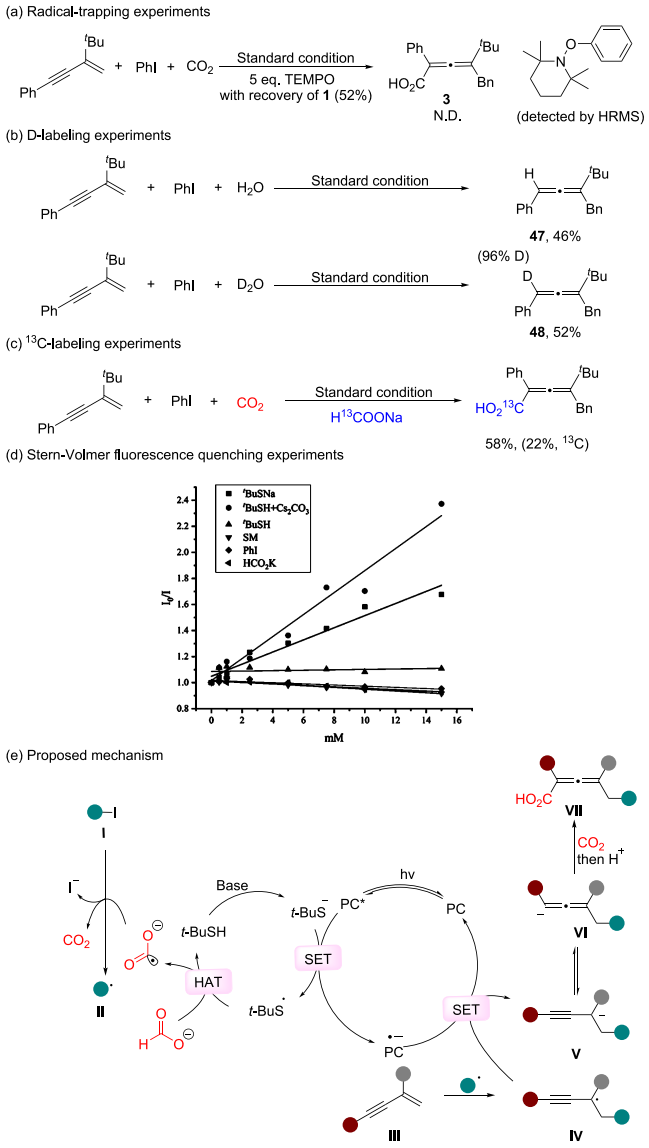
(Scheme 3b), **3** reacted with methoxy methylamine hydrochloride to furnish Weinreb amide **42** in 90% yield. Subsequent treatment of **42** with methylmagnesium bromide generated allenone **43**. Silver nitrate- catalyzed cyclization of **3** proceeded to give furanone **44**, featuring an all-carbon quaternary center in 97% yield. Furthermore, **3** underwent one-step esterification with $TMSCHN_2$ to deliver allenoate **45** in 99% yield, which was conveniently converted to allenol **46** via diisobutylaluminum hydride (DIBAL-H) reduction.

To elucidate the reaction mechanism, 2,2,6,6-tetramethylpiperidin-1-oxyl (TEMPO) as a radical scavenger was introduced into the standard reaction system (Scheme 4a). The reaction was completely suppressed with 52% recovery of substrate **1**, and the TEMPO adduct was detected by HRMS, suggesting a potential radical pathway. When the reaction was conducted under N_2 atmosphere with added H_2O, allene product **47** was obtained in 46% isolated yield. Employing D_2O afforded deuterated product **48** in 52% yield with 96% deuterium incorporation (Scheme 4b), directly confirming the existence of an allene carbanion intermediate. To elucidate the carboxyl source, $H^{13}COONa$ was employed as a tracer, and 22% ^{13}C incorporation was observed at the carboxyl group of the product, indicating that formate also serves as a carbon source for carboxylation (Scheme 4c). Additionally, Stern−Volmer quenching experiments (Scheme 4d) further demonstrated that thiolate effectively quenches the excited state of the photocatalyst.

Based on these results and literature evidence,[13a,m] a plausible reaction mechanism is proposed (Scheme 4e). Photoexcitation of photocatalyst PC generates excited-state PC*, which undergoes single-electron reduction quenching by

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Scheme 4. Control Experiments and Proposed Mechanism



thiolate to produce reduced $PC^{\bullet-}$ and a thiyl radical. This thiyl radical then activates formate via hydrogen atom transfer (HAT), yielding the key reactive species $CO_2^{\bullet-}$. Subsequent reduction of iodide **I** by $CO_2^{\bullet-}$ forms carbon radical **II**, which adds to enyne **III** affording propargyl radical **IV**, followed by reduction (via $PC^{\bullet-}$ or $CO_2^{\bullet-}$) to a carbanion intermediate **V**. Isomerization of **V** generates allene carbanion **VI** and carboxylation with CO_2, final protonation delivers product **VII**.

In summary, we have developed an efficient photocatalytic 1,4-carbocarboxylation of 1,3-enynes with CO_2 to construct tetrasubstituted 2,3-allenoic acids under mild conditions. Key to this strategy is the synergistic combination of thiol HAT catalysis and formate, which concurrently generates reactive $CO_2^{\bullet-}$ radicals. The method exhibits exceptional functional group tolerance and substrate generality, accommodating diverse (hetero)aryl and alkyl iodides, and enabling late-stage modification of pharmaceuticals. This work establishes a sustainable approach to valuable allene carboxylic acid architectures using formate and CO_2 as synergistic carbon sources.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c03750.

Experimental procedures, spectra for all new compounds (PDF)

Accession Codes

Deposition Number 2473796 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Author Contributions

§D.-S.D. and F.C. contributed equally to this work.

Notes

The authors declare no competing financial interest.

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